

Chemical Engineering for Humanity

CHEN 1000, Fall 2025

Project Proposal. You now have the opportunity to propose a new project to evaluate how chemical engineers can contribute to humanity. The proposal should follow the format provided in the three assigned projects this semester – this includes an introductory framing and several potential questions. You should identify something you wish you had learned more about in this introductory seminar related to a major technical challenge that chemical engineers can help solve and propose a project to investigate more about this topic. Green hydrogen? Decarbonization of steel manufacturing? Production of bioengineered meats? Utilization of agricultural waste for materials? Scale-up of personalized medicine? Production of gene therapies? Etc.

1. What is the motivation for the project? What is the big picture question that should be investigated?
2. What are one or two calculations or process diagrams that can be performed related to the proposed project area?
3. What are economic, societal, environmental, cultural trade-offs of the proposed technical solution that should be considered?

Your completed proposal should include the following:

- A 250-300 word project introduction with 2-4 questions related to the project. These questions should include at least 1 basic calculation or process flow diagram.
- You should also provide the project proposal as a prompt to a generative AI algorithm (e.g. use LibreChat) and evaluate the answers it provides. On a separate page from your project proposal, you should turn in the prompt, the genAI response(s), and your evaluation of the responses.